

MICAH OEVERMANN, PHD

Texas A&M University ◇ College Station, Texas ◇ U.S. Citizen

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EDUCATION

PhD in Mechanical Engineering

December 2025

Texas A&M University, College Station, USA

Dissertation Title: Design, Modeling, and Nutational Instabilities of Soft Pendulum Driven Spherical Robots

B.S. in Mechanical Engineering

December 2021

Texas A&M University, College Station, USA

Study Abroad - Engineering Mechanics

Summer 2019

Arts et Métiers ParisTech, Aix-en-Provence, France

EXPERIENCE

Robotics and Automation Design Lab

January 2022 - December 2025

Graduate Research Assistant

College Station, TX

- Led a small team of grad students and engineers on the full-stack development of the RoboBall II prototypes from CAD assemblies in Solidworks to electrical wiring and crimping of components through weekly design reviews
- Patenting a novel outer shell manufacturing method used in the two and six-foot RoboBalls
- Led the control system design for the ball in Wpilib c++ and later ros2 using input filtering and LQR feedback regulation
- Iterated on the above bullets through constant field testing and redesign of the system.
- Led teams of 5, 3, and 2 undergraduates in the three summer programs, all resulting in 3 conference and a journal publications
- Collaborated with the Robotic Space Simulator team on the motion control two 7-dof Stewart platforms with UR-20 robotic arms
- Collaborated with external partners at Southwest Research Institute for the development of motion planning libraries
- authored updated versions of journal L^AT_EX templates to make lab publishing more streamlined

BakerRisk Engineering Consultants

August 2020 - December 2020

Student Co-op, Blast Testing Group

San Antonio, TX

- Destructive full-scale structural testing with Deflagration, Vapor Cloud, and Shock Tube methods
- Manufactured mounts for window specimens and piezoelectric pressure or force instrumentation
- Prioritized safety with no major injuries while working around debris fields with broken glass

Biomechanical Environments Laboratory

January 2019 - May 2019

Undergrad Research Assistant

College Station, TX

- Applied concepts of linear elastic theory in the development of a biaxial tissue testing platform
- Implemented the use of a novel fish hook – line technique on organic samples to reduce clamp stresses
- Presented final poster in a public research symposium

TEACHING AND COUNSELING EXPERIENCE

Conference Paper Reviewer

March 2025-October 2025

- UR 2025 and ICRA 2026 Conferences

Guest Speaker - MEEN 612

December 2024

Introduction to Robotic Manipulators

College Station, TX

- Lecture Title: *Introduction to Simulating Robot Arms in pyDrake*

Guest Speaker - MEEN 689

December 2024

Intuitive and Counterintuitive Mechanisms

College Station, TX

- Lecture Title: *Designing for Assembly: Lessons from RoboBall II Pendulum*

Grad Camp Counselor

August 2022 - September 2022

Graduate Orientation Camp

College Station, TX

- Assist incoming graduate students at introductory sessions to the university

Numerical Methods Helpdesk

August 2019 - December 2019

Student Worker for MEEN 357

College Station, TX

- Assist students with Python code portions of projects and homework

Volunteer Teaching Assistant

August 2018 - December 2018

Kinesiology 199 - Whitewater Kayaking

College Station, TX

- One-on-one instruction and feedback on fundamental paddling techniques

Python Teaching Assistant

August 2018 - December 2018

Student Assistant for Intro Engineering Course ENGR 102

College Station, TX

- Assisted in the instruction of freshman engineering students with the basics of coding in Python

FUNDING AND SCHOLARSHIPS

Lou & CC Burton '42 Scholarship ◇ \$10,000

Awarded 2020-2021

Lechner Scholarship ◇ \$10,000

Awarded 2017-2020

CITGO Petroleum Corp. Scholarship

Awarded 2018-2019

Pat Wilburn Honorary Scholarship

Awarded 2017-2019

HONORS AND AWARDS

Best Presentation Award

September 2023

OSU International Mechatronics Conference and Exposition, 2023

Senior Capstone: Best in MEEN

December 2021

Texas A&M University, College Station, USA

TECHNICAL SKILL BUZZWORDS

Mechanical: SolidWorks ◇ Rapid Prototyping ◇ Machining ◇ Mechanism Design ◇ GD&T ◇ Dynamic Analysis ◇ Molding ◇ Hermetic Seals

Software: C++ ◇ Python ◇ MATLAB ◇ ROS2 ◇ Drake ◇ Linux ◇ Git/Github ◇ Docker ◇ LLM Support ◇ Energy Methods ◇ L^AT_EX

Electrical: Soldering ◇ Oscilloscopes ◇ Sensor Integration ◇ Cable Harness ◇ CAN/Ethernet Bus ◇ Battery Management Systems

Controls: PID ◇ Disturbance Rejection ◇ LQR ◇ Full-State Feedback Motion Planning ◇ State-Space Control ◇ Underactuated Systems

PATENT APPLICATIONS

Interior Molding Method for Construction of configurable Bead Locked Spherical Tires

U.S. Provisional Application No.: 63/940,051 *Filing Date: December 13, 2025*

Journal Articles

Novel Robotic Fleet for Sample Recovery in Lunar Craters: A Concept of Operations

Jangale, R., D. Pravecek, S. Lam, D. McDougall, M. Treviño, A. Villanueva, J. Land, H. De Leon, **Oevermann, M.**, and R. Ambrose

IEEE Transactions on Field Robotics 2 (2025) pp. 937–955. 2025

Empirical Contact Models for Soft Spherical Robots in Drake

Oevermann, Micah J., Dhruv Datta, Dylan Hilburn, Derek J. Pravecek, Rishi Jangale, Aaron Villanueva, and Robert O. Ambrose

IEEE Robotics and Automation Letters (2025). 2025

Empirically Compensated Setpoint Tracking for Spherical Robots With Pressurized Soft-Shells

Pravecek, Derek J, **Micah J Oevermann**, Gray C Thomas, and Robert O Ambrose

IEEE Robotics and Automation Letters (2025). 2025

Peer Reviewed Conference Papers

Scaling of RoboBall: A Parametric Robot Family for Crater Exploration

Jangale, Rishi V, Aaron Villanueva, Garrett Jibrail, **Micah J Oevermann**, Derek J Pravecek, Meghali P Dravid, and Robert O Ambrose

2025 IEEE Aerospace Conference, 2025

A Pressure Model and Control System for a Pressurized Pendulum Driven Spherical Robot

Micah J Oevermann, Meghali P Dravid, Derek J Pravecek, Will Olejnik, and Robert O Ambrose

2025 22nd International Conference on Ubiquitous Robots (UR), 2025

Design of a Soft Shell for a Spherical Exploration Robot Traversing Varying Terrain

Dravid, Meghali Prashant, **Micah Oevermann**, David McDougall, David Dugas, and Robert Ambrose

2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024

A Soft Spherical Robot for Lunar Crater Exploration

Micah Oevermann, Meghali Prashant Dravid, Garrett Jibrail, Jared Janak, Rishi Jangale, David McDougall, David Dugas, and Robert O Ambrose

AIAA SCITECH 2024 Forum, 1961, 2024, 2024

Roboball: An all-terrain spherical robot with a pressurized shell

Micah Oevermann, Derek Pravecek, Garrett Jibrail, Rishi Jangale, and Robert O Ambrose

2024 IEEE International Conference on Robotics and Automation (ICRA), 2024

Presented Abstracts

A System for Exploring Craters and Shadowed Regions of the Lunar South Pole

Dravid, Meghali, **Micah Oevermann**, and Robert Ambrose

ASCE Space and Earth Conference, 2024

RoboBall Recap: Past, Current, and Future Inflatable Spherical Robots

Jangale, Rishi, **Micah Oevermann**, Garrett Jibrail, Derek Pravecek, Meghali Dravid, Aaron Villanueva, and Robert Ambrose

40th Anniversary of the IEEE International Conference on Robotics and Automation, 2024

Persistent intelligence, Surveillance and Reconnaissance for the Lunar Surface

Ambrose, Robert, **Micah Oevermann**, Meghali Dravid, and Garrett Jibrail

AIAA ASCEND Conference, 2023

Design and Dynamics of Rugged Soft Shells for a Pendulum-Driven Spherical Robot

Micah Oevermann, Meghali Dravid, Garrett Jibrail, and Robert Ambrose

OSU International Mechatronics Conference and Exposition, 2023

Works Under Review

Relaxation Dynamics in Oblate Spherical Rolling Robots

Oevermann, Micah J. and Robert O. Ambrose

Proceedings of the IEEE International Conference on Robotics and Automation (ICRA) (2026). 2026